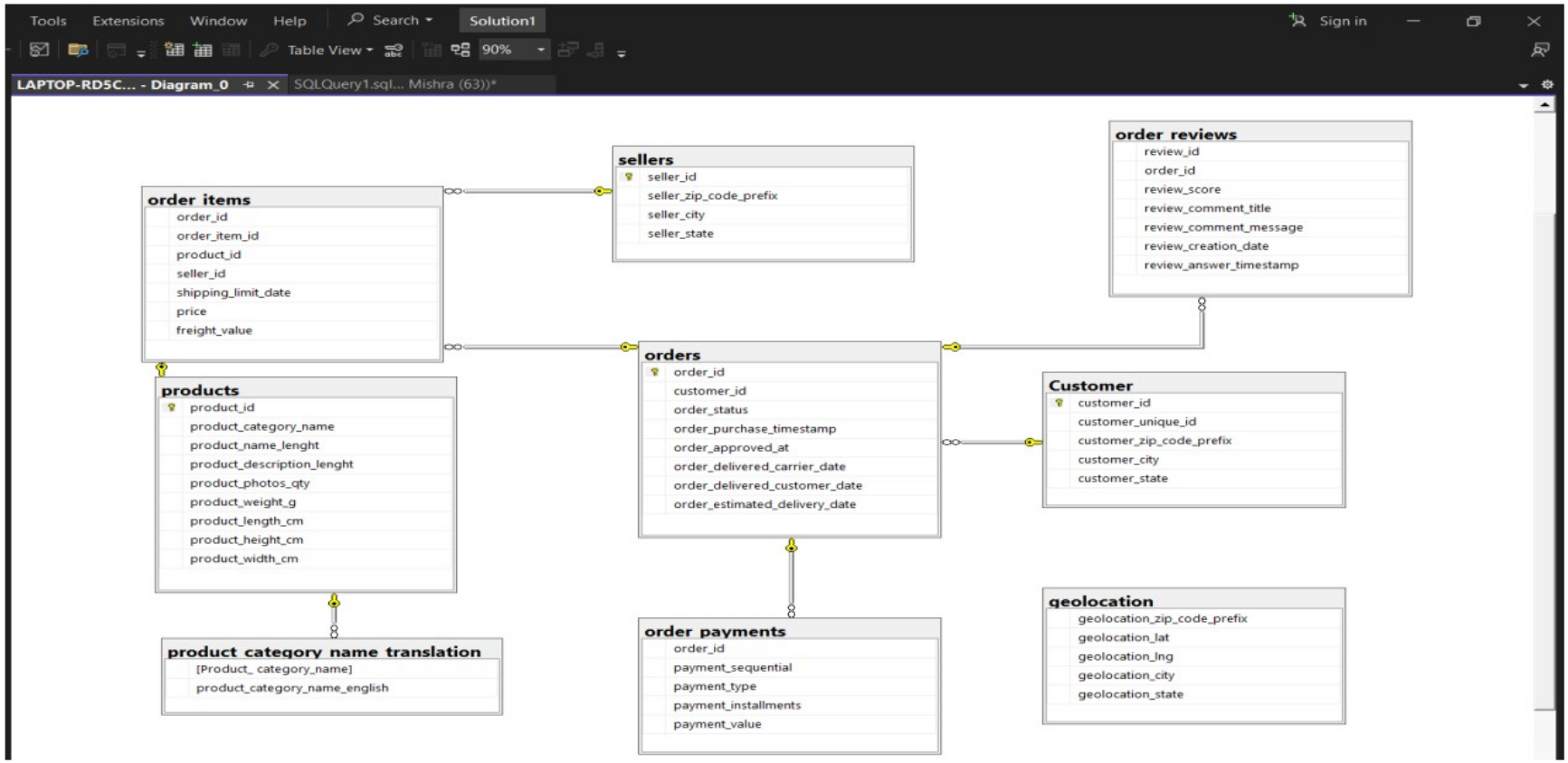
Entity-Relationship Diagram



Available Data

The Olist Brazilian E-Commerce dataset consists of 9 relational tables containing information about customers, orders, products, sellers, payments, reviews, and geolocation.

customers

Stores customer information and location

PK customer_id

orders

Contains order details, status, and timestamps

PK order_id

order_items

Details of products included in each order

PK order_id + order_item_id

products

Product information and attributes

PK product_id

sellers

Seller information and location

PK seller_id

order_payments

Payment details for each order

PK order_id

order_reviews

Customer review scores and comments

PK review_id

geolocation

Geographic data based on ZIP codes

PK geolocation_zip_code_prefix

product_category_name_translation

Maps Portuguese categories to English

PK product_category_name

DATA ARCHITECTURE — DATA DICTIONARY

products

COLUMN	TYPE	DESCRIPTION
PK **product_id**	NVARCHAR(50)	Unique product identifier
product_category_name	NVARCHAR(50)	Product category
product_name_length	INTEGER	Product name length
product_description_length	INTEGER	Description length
product_photos_qty	INTEGER	Number of product images
product_weight_g	INTEGER	Product weight (grams)
product_length_cm	INTEGER	Product length (cm)
product_height_cm	INTEGER	Product height (cm)
product_width_cm	INTEGER	Product width (cm)

order_items

COLUMN	TYPE	DESCRIPTION
PK **order_id**	NVARCHAR(50)	Order identifier
PK **order_item_id**	INTEGER	Item sequence within an order
FK **product_id**	NVARCHAR(50)	Purchased product
FK **seller_id**	NVARCHAR(50)	Seller of the product
shipping_limit_date	DATETIME	Shipping deadline
price	FLOAT	Product price
freight_value	FLOAT	Shipping cost

sellers

	COLUMN	TYPE	DESCRIPTION
PK	**seller_id**	NVARCHAR(50)	Unique seller identifier
	seller_zip_code_prefix	INTEGER	Seller ZIP code
	seller_city	NVARCHAR(50)	Seller city
	seller_state	NVARCHAR(50)	Seller state

order_payments

	COLUMN	TYPE	DESCRIPTION
FK	**order_id**	NVARCHAR(50)	Order identifier
PK	**payment_sequential**	INTEGER	Payment sequence number
	payment_type	NVARCHAR(50)	Payment method
	payment_installments	INTEGER	Number of installments
	payment_value	FLOAT	Payment amount

order_reviews

	COLUMN	TYPE	DESCRIPTION
PK	**review_id**	NVARCHAR(50)	Unique review identifier
FK	**order_id**	NVARCHAR(50)	Associated order
	review_score	INTEGER	Customer rating (1-5)
	review_comment_title	NVARCHAR(50)	Review title
	review_comment_message	NVARCHAR(50)	Customer review
	review_creation_date	DATETIME	Review creation date
	review_answer_timestamp	DATETIME	Review response date

geolocation

COLUMN	TYPE	DESCRIPTION
PK geolocation_zip_code_prefix	INTEGER	ZIP code prefix
geolocation_lat	FLOAT	Latitude
geolocation_lng	FLOAT	Longitude
geolocation_city	NVARCHAR(50)	City
geolocation_state	NVARCHAR(50)	State

orders

COLUMN	TYPE	DESCRIPTION
PK **order_id**	NVARCHAR(50)	Unique order identifier
FK **customer_id**	NVARCHAR(50)	Customer who placed the order
order_status	NVARCHAR(50)	Current order status
order_purchase_timestamp	DATETIME	Order purchase date and time
order_approved_at	DATETIME	Order approval time
order_delivered_carrier_date	DATETIME	Shipment date
order_delivered_customer_date	DATETIME	Delivery date
order_estimated_delivery_date	DATETIME	Estimated delivery date

product_category_name_translation

COLUMN	TYPE	DESCRIPTION
PK **product_category_name**	NVARCHAR(50)	Product category (Portuguese)
product_category_name_english	NVARCHAR(50)	Product category (English)

customers

COLUMN	TYPE	DESCRIPTION
PK **customer_id**	NVARCHAR(50)	Unique customer identifier
customer_unique_id	NVARCHAR(50)	Unique identifier for repeat customers
customer_zip_code_prefix	INTEGER	Customer ZIP code prefix
customer_city	NVARCHAR(50)	Customer city
customer_state	NVARCHAR(50)	Customer state